

SATNET EDU / Lecture 01

Satellite Network Architecture

How orbit, coverage, and routing shape a packet's journey.

LECTURE 01

An introduction to low Earth orbit (LEO) networks

Trace the path. Explain the constraints. Predict what changes.

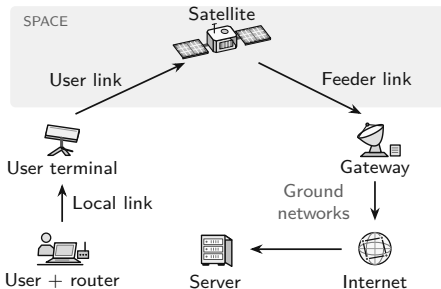
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From the User Terminal to the Internet

Follow one outbound packet from a user device, through space, to an Internet server.

- ▶ **The terminal provides radio access.**
 - A router feeds the terminal. The **user link** then carries traffic from terminal to satellite.
- ▶ **The gateway reconnects to Earth.**
 - Its **feeder link** carries traffic from space. Ground networks deliver the packet to the server.
- ▶ **Every part of the path must work.**
 - A visible satellite alone is insufficient. Radio links, the gateway, and the onward route must all be usable.

One packet's complete access path



Generic single-satellite access route. Replies require a return route, which may be different.

Low Orbits Trade Shorter Paths for Motion

Orbit altitude changes both signal travel time and how a satellite moves relative to a user.

► **Short paths reduce travel time.**

- For one vertical leg, distance equals altitude. At 550 km, the signal takes about 1.8 ms to reach the satellite.

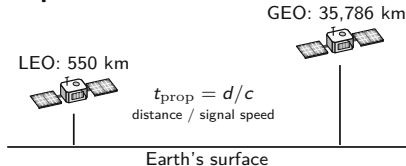
► **LEO satellites move across the sky.**

- A 550 km circular orbit takes 95.5 minutes at about 7.6 km/s. A fixed terminal must change satellites over time.

► **A GEO satellite stays in one place.**

- A geostationary satellite stays above one equatorial point, but has a much longer radio path.

Compare one ideal vertical radio leg



	550 km LEO	GEO
One vertical leg	1.8 ms	119.3 ms
Orbit period	95.5 min	23 h 56 min
Position in sky	Changes	Fixed

Circular orbits. Earth radius 6371 km, signal speed $c \approx 300,000$ km/s. The sketch is not to scale.

Elevation Determines Usable Contacts

Elevation is the angle above the local horizon. Altitude alone does not give link distance.

► **Lower elevation gives a longer path.**

- The **slant range** is the straight-line distance to the satellite. It increases as elevation falls.

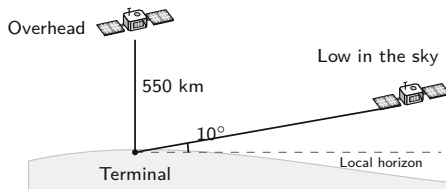
► **A mask removes low-angle contacts.**

- Low angles face more blockage and atmosphere. A minimum elevation, such as 25° , is a design choice.

► **A higher mask reduces coverage.**

- It shrinks the usable ground region and shortens the time a user can stay connected to one satellite.

The same altitude, different slant ranges



Elevation	Slant range	One-way time
90°	550 km	1.8 ms
30°	993 km	3.3 ms
10°	1815 km	6.1 ms

Spherical Earth, radius 6371 km, altitude 550 km. Table values use spherical geometry, not the sketch's scale.

Constellations Spread Coverage Across Earth

A constellation coordinates many satellites across several orbital planes.

► **Planes spread satellites globally.**

- Satellites follow one another within a plane. A **shell** groups orbits with similar altitude and tilt.

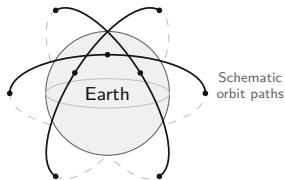
► **Inclination sets latitude reach.**

- The orbit's tilt relative to the equator limits the latitudes reached by the point directly below the satellite.

► **Phasing helps fill coverage gaps.**

- Offsets stagger positions between planes. New satellites must arrive as others leave, with usable links and capacity.

Three planes, four satellites per plane



Unwrapped positions within each plane



$$3 \text{ planes} \times 4 \text{ satellites} = 12 \text{ satellites}$$

Illustrative shell. Ground coverage extends away from each satellite's ground track. This is not a service-coverage map.

Coverage Does Not Guarantee Capacity

Radio resources are divided among beams and among the active users inside each beam.

► **Spot beams focus radio energy.**

- Different beams serve different areas.
Reusing spectrum needs isolation or interference control.

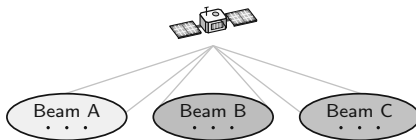
► **Beam capacity is shared by users.**

- The scheduler allocates time, frequency, and power. More active users can reduce each user's share.

► **Any link can become a bottleneck.**

- A strong user link cannot overcome a congested feeder link, a slow space path, or a limited ground connection.

Coverage and shared resources



Dots represent users in each beam

Equal sharing of one 100 Mb/s beam

4 active users: $100/4 = 25$ Mb/s each

A	B	C	D
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8 active users: $100/8 = 12.5$ Mb/s each

A	B	C	D	E	F	G	H
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Teaching example: equal saturated users, enough onward capacity, and no protocol overhead. Real schedulers need not share equally.

Inter-Satellite Links Extend the Space Route

Traffic can return to Earth after one satellite or travel farther through space.

► **A relay needs two ground contacts.**

- The user terminal and a connected gateway must both reach the same satellite.

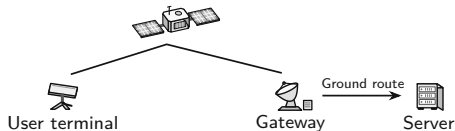
► **ISLs create a route through space.**

- An **inter-satellite link** uses radio or optical terminals. A chain can reach a gateway beyond the first satellite's view.

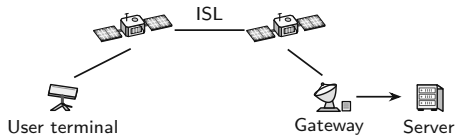
► **Gateway choice changes the route.**

- Extra space hops may avoid a ground detour, but add distance and link requirements. Fewer hops need not mean less delay.

One satellite, then ground networks



A route with inter-satellite links



Generic architectures. ISLs need line of sight, suitable terminals, and accurate pointing. Neither route is always faster.

Handover Uses Overlapping Contact Windows

A fixed terminal can lose its serving satellite even when neither endpoint has failed.

► **Every pass has a finite window.**

- Elevation rises and falls. Below the minimum angle, or mask, a satellite is no longer usable in this model.

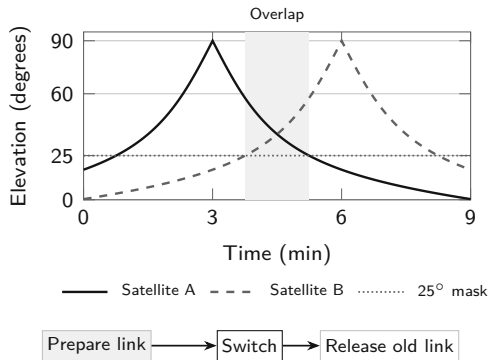
► **Overlap gives time to prepare.**

- The next link and onward route should be ready before the current contact ends. Here, overlap lasts from 3.8 to 5.2 min.

► **Visibility cannot ensure continuity.**

- Terminal capability, signaling, and resource allocation determine whether traffic continues or experiences a gap.

Two passes, one opportunity to switch



Illustrative overhead passes at 550 km, peaking at 3 and 6 min. Earth rotation and handover overhead are omitted.

Internet Latency Includes More Than Propagation

An end-to-end packet crosses several links and waits or is processed at intermediate nodes.

► **Signals take time to cross each link.**

- Radio travels at about 300,000 km/s in space. Light in fiber travels at about 200,000 km/s.

► **Packets also spend time at nodes.**

- Transmission takes packet length / link rate. Processing and queues add delay. Queueing varies with load.

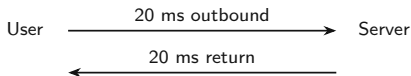
► **RTT includes both directions.**

- Round-trip time is about twice the one-way delay only when both directions have similar paths and delays.

End-to-end delay budget

Contribution	What it counts	Total
Propagation	Distance / signal speed	12 ms
Transmission	Packet length / link rate	1 ms
Processing	Work done at nodes	1 ms
Queueing	Wait before transmission	6 ms

One way: $12 + 1 + 1 + 6 = 20$ ms



Symmetric example: $RTT \approx 40$ ms

Illustrative packet budget. Each contribution is already summed across the path. These are not measured service rates or delays.

Model the Network as a Changing Graph

At one instant, build the usable links first. Then choose a complete route through them.

► **Geometry determines usable links.**

- Nodes are satellites and ground endpoints. Links need visibility, range, and modeled equipment conditions.

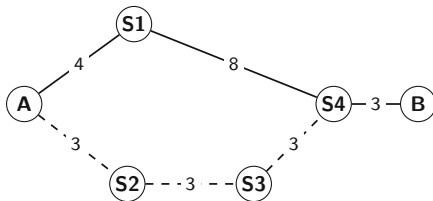
► **The objective ranks complete paths.**

- Minimum hops counts links. Minimum propagation delay adds link delays. They can select different paths.

► **Time changes the graph and routes.**

- Motion or an outage can remove links. SatNet Edu models these changes, but omits radio capacity, queues, and handover signaling.

Fewer hops can still mean more delay



Link weights: one-way propagation delay in ms

Path	Hops	Delay
A-S1-S4-B	3	$4 + 8 + 3 = 15$ ms
A-S2-S3-S4-B	4	$3 + 3 + 3 + 3 = 12$ ms

Illustrative graph. A and B are ground endpoints, S1-S4 are satellites. Weights are a teaching example, not a physical constellation.